ENGINEERING & SCIENCE STUDENT DESIGN SHOWCASE

FLORIDA TECH

HALO: High Autonomous Low-SWaP Operations

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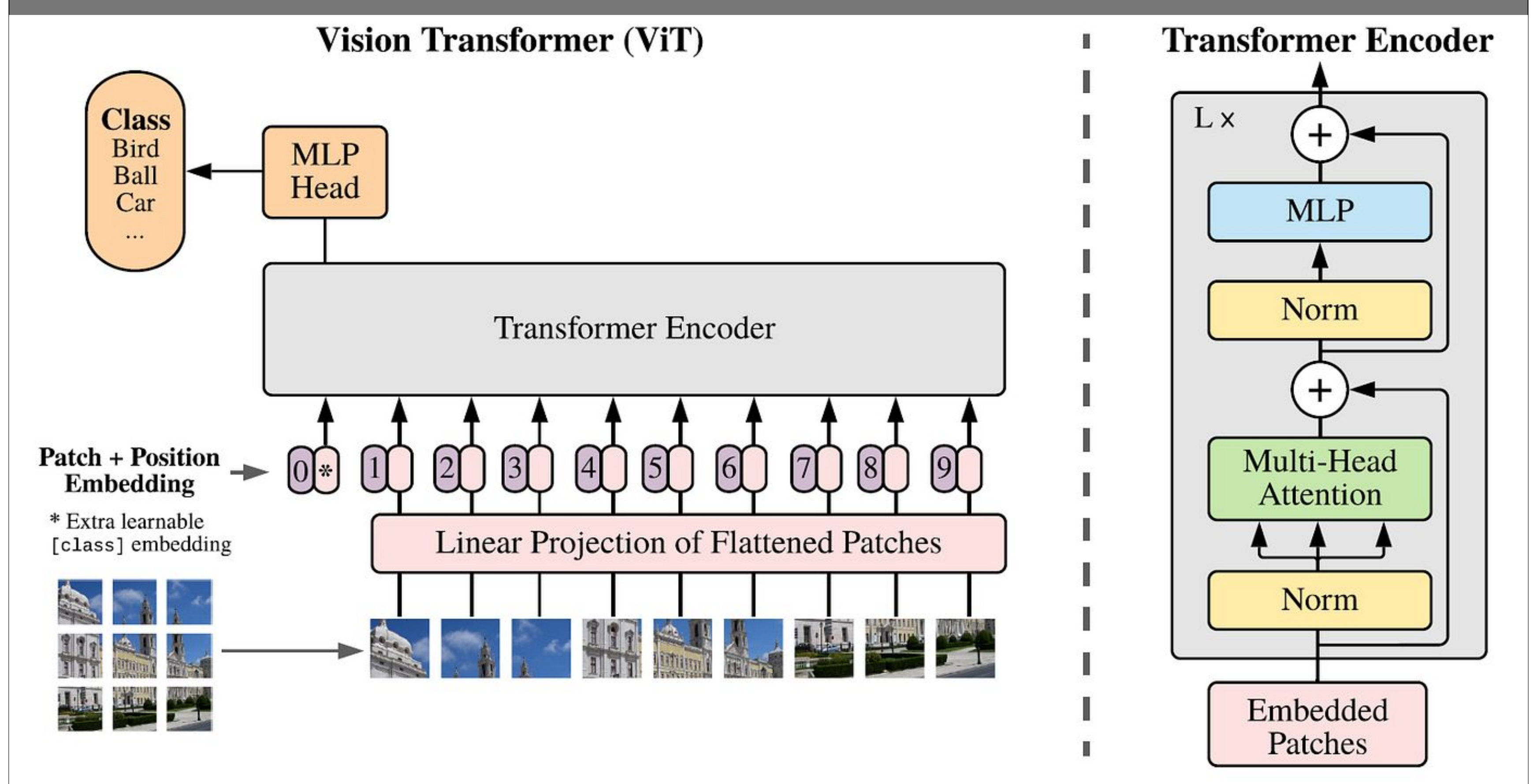
ABSTRACT

Orbital object detection is a vital aspect of space operations. More specifically, it is important and extremely helpful when attempting to identify satellite components. Space-based object detection is typically achieved through ground-based radars, optical telescopes, and sensors fixed to satellites; however, as the amount of space debris grows, it is becoming more difficult to use those systems for object detection. Additionally, the need for real-time orbital object detection is growing, due to the previously listed factors. In order to achieve real-time orbital object detection, Machine Learning comes into play. Neural networks, typically convolutional neural networks (CNNs), are used by running on-board models directly on satellite systems. For this project, we employed the use of a Vision Transformer (ViT), a deep learning model handles vision processing tasks. Often, the problem that engineers and scientists tend to run into when deploying systems with such capabilities is resource allocation. Neural network models are typically run on large computers, which take up lots of space, power, and other resources on an already limited system. A solution to this is to scale down the computer to a Low-SWaP system (Low-SWaP meaning low size, weight, and power); however, that comes at the cost of a large model running inefficiently on a small computer due to limited memory and less power draw. In order for a ViT to run efficiently on a smaller computer, one must shrink the model itself; this is done through a process called quantization. Quantization converts high-precision numerical values to lower-precision representations, which reduces a models memory footprint, accelerates inference speed, and lowers energy consumption, ultimately enabling efficient deployment on edge devices without significant accuracy loss. A fully trained Vision Transformer neural network model is represented in 32-bit; a ViT can be quantized down to 16-, 8-, 4-, novelly 2-, and theoretically 1-bit. The project objective is to quantize a Vision Transformer model and report its performance at each of these end points. This process will allow us to reach our goal of achieving real-time orbital object detection by deploying a Vision Transformer onto a Low-SWaP computer; colloquially termed as achieving High Autonomous Low-SWaP Operations, or HALO.

MOTIVATION

METHODS

- Dr. White's NEural TransmissionS (NETS) Lab GPU servers for training the 32-bit Vision Transformer
- Python
- NVIDIA Jetson AGX Orin for ViT deployment
- NVIDIA TensorRT SDK
- Quantization techniques, including Post Training Quantization (PTQ), and Quantize/Dequantize (Q/DQ) Node insertion for Quantized Aware Training (QAT)



Images of Dataset

Images of Detections

NOVELTY

Vision Transformers (ViTs) are a relatively new idea in the world of Neural Networks and Machine Learning, and, even further, so is quantizing them to run efficiently on Low-SWaP computers. Orbital object detection is typically done through convolutional neural networks (CNNs); however, ViTs are better suited for such tasks. Vision Transformers offer the ability to capture global context, meaning a model is able to discern relationships between distant objects or features within a vast, complex image; while CNNs are better suited for identifying small objects in isolation rather than capturing the broader context of satellite imagery. Further, quantization, while not a new concept in deep learning, is a relatively recent, growing area of research when applied to Vision Transformers. Quantizing ViTs proves uniquely challenging due to severe activation outliers and sensitive attention mechanisms, requiring specialized methods to minimize accuracy loss.

RESULTS

FUTURE WORK

- Achieving a true 2-bit representation of a Vision Transformer for orbital object detection tasks
- Achieving a true 1-bit representation of a Vision Transformer for orbital object detection tasks
- Efficient satellite component object detection for:
  - Diagnosing broken components
  - Docking onto satellites
  - Identifying space debris

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